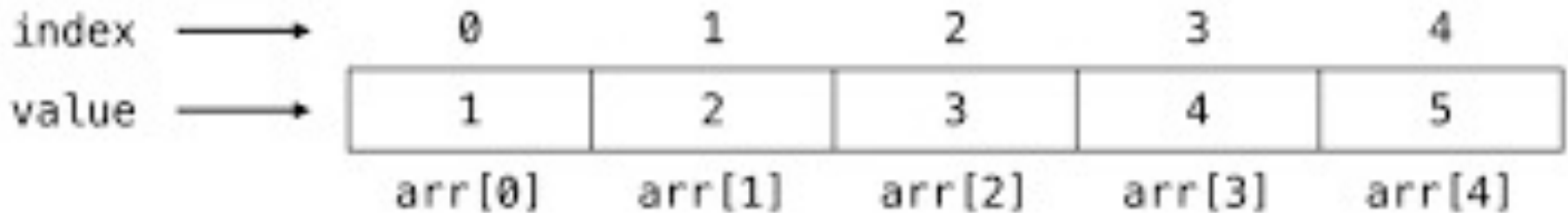


2-Dimensional (2D) Array

What is 1D Array?

- ❖ A 1-dimensional (1D) array is a **linear list** where each element is **accessed** using **column index**.

```
int[] arr = {1, 2, 3, 4, 5};
```



What is 2D Array?

- ❖ A 2-dimensional (2D) array is a **matrix** where each element is **accessed** using both **row** and **column index**.

2 D ARRAY:

		col 0	col 1	col 2	
	i \ j	0	1	2	← column
row 0	0	A	A	A	} array elements
row 1	1	B	B	B	
row 2	2	C	C	C	

↑
ROWS

Some sample parts of 2D programming

- ❖ Taking Input
- ❖ Printing Input
- ❖ Some matrix operations like addition, multiplication and transpose.

Taking input in 2D Array

- ❖ Taking input in every row
- ❖ For each row, taking input in each column

Declaration of 2D Array

```
int A[100][100];
```

Say, N = number of columns

M = number of rows

For square matrix, N = M

```
for ( i = 0; i < M; ++i)
    for ( j = 0; j < N; ++j){
        printf("Enter value A[%d][%d]: ", i+1, j+1);
        scanf("%d", &A[i][j] );
    }
```

Printing Output in 2D Array

- ❖ Printing each row
- ❖ For each row, printing each column

Declaration of 2D Array

```
int A[100][100];
```

Say, N = number of columns

M = number of rows

For square matrix, N = M

```
for ( i = 0; i < M; ++i) {  
    for ( j = 0; j < N; ++j)  
        printf(“%4d“, A[i][j] );  
    printf(‘\n’);  
}
```

Matrix (2D Array) Addition

$$\begin{bmatrix} 12 & 9 & 7 \\ 5 & 11 & 8 \\ 15 & 3 & 2 \end{bmatrix} + \begin{bmatrix} 8 & 6 & 13 \\ 6 & 14 & 9 \\ 5 & 13 & 7 \end{bmatrix} = \begin{bmatrix} 12+8 & 9+6 & 7+13 \\ 5+6 & 11+14 & 8+9 \\ 15+5 & 3+13 & 2+7 \end{bmatrix}$$

A **B** **C**

Declaration of 2D Array

```
int A[10][10], B[10][10], C[10][10];
```

```
for ( i = 0; i < M; ++i)
    for ( j = 0; j < N; ++j)
        C[i][j] = A[i][j] + B[i][j];
```

Complete Program

```
#include <stdio.h>

main(){
    int A[10][10], B[10][10], C[10][10];
    int i, j, N, M;

    printf("How many rows?: ");
    scanf("%d", &M);
    printf("How many columns?: ");
    scanf("%d", &N);

    for ( i = 0; i < M; ++i)
        for ( j = 0; j < N; ++j){
            printf("Enter value A[%d][%d]: ", i+1, j+1);
            scanf("%d", &A[i][j] );
        }
}
```

Complete Program continue...

```
for ( i = 0; i < M; ++i)
    for ( j = 0; j < N; ++j){
        printf("Enter value B[%d][%d]: ", i+1, j+1);
        scanf("%d", &B[i][j] );
    }
```

```
for ( i = 0; i < M; ++i)
    for ( j = 0; j < N; ++j)
        C[i][j] = A[i][j] + B[i][j];
```

```
for ( i = 0; i < M; ++i) {
    for ( j = 0; j < N; ++j)
        printf("%4d", C[i][j] );
    printf('\n');
}
}
```

Matrix (2D Array) Multiplication

$$\begin{bmatrix} 12 & 9 & 7 \\ 5 & 11 & 8 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 8 & 6 & 13 \\ 6 & 14 & 9 \\ 5 & 13 & 7 \end{bmatrix} = \begin{bmatrix} C[0][0]=185 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

A **B** **C**

$$C[0][0] = 12 \times 8 + 9 \times 6 + 7 \times 5 = 96 + 54 + 35 = 185$$

$$\begin{bmatrix} 12 & 9 & 7 \\ 5 & 11 & 8 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 8 & 6 & 13 \\ 6 & 14 & 9 \\ 5 & 13 & 7 \end{bmatrix} = \begin{bmatrix} C[0][0]=185 & C[0][1] & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

A **B** **C**

$$C[0][1] = 12 \times 6 + 9 \times 14 + 7 \times 13$$

Matrix (2D Array) Multiplication

- If matrix A has a dimension of $M \times N$ and B has a dimension of $N \times P$, then resultant matrix will be $M \times P$.
- For $A \times B$, the number of column of matrix A must be the number of row of matrix B.

```
for ( i = 0; i < M; ++i)
    for ( j = 0; j < P; ++j){
        C[i][j] = 0;
        for ( k = 0; k < N; ++k)
            C[i][j] += A[i][k] * B[k][j];
    }
```

Matrix (2D Array) Transpose

$$A = \begin{bmatrix} 12 & 9 & 7 \\ 5 & 11 & 8 \\ 15 & 3 & 2 \end{bmatrix}$$

$$A' = \begin{bmatrix} 12 & 5 & 15 \\ 9 & 11 & 3 \\ 7 & 8 & 2 \end{bmatrix}$$

- No change in diagonal values, when $i = j$.
- For other values, swap $A[i][j]$ and $A[j][i]$.

```
for ( i = 0; i < N; ++i)
    for ( j = 0; j < i; ++j){
        temp = A[i][j];
        A[i][j] = A[j][i]
        A[j][i] = temp;
    }
```